

Modelling and Numerical Study of Mg-CO₂ Electrochemical System for CO₂ Utilization with Electricity and Hydrogen Production

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After industrialization, the world's atmospheric carbon dioxide, a greenhouse gas, content increased from 287 ppm to 417 ppm because of the increased demand for energy. It affects the environment in many ways, including sea level rise, unpredictable floods and droughts, and global warming [1]. Several chemical and electrochemical carbon capture, utilization, and storage and sequestration (CCUS) technologies are being developed to reduce CO₂ concentration, but they are still inadequate due to low conversion efficiency and economic feasibility [2].

Electrochemically, CO₂ conversion into a useful product or fuel has the potential to reduce CO₂ emissions worldwide. In this study, a 3D multiphase transient Mg - CO₂ electrochemical model based on water and CO₂ reactions is developed. This system is able to convert CO₂ into Mg(HCO₃)₂, H₂ (a green fuel), and electricity. CO₂ gas typically forms H⁺, HCO₃⁻ and CO₃²⁻ in aqueous electrolyte. The presence of H⁺ ions leads to a decrease in the pH of the electrolyte, making the solution acidic. This acidity of water is utilized to generate H₂ at the cathode [3]. The model consists of a Mg metal anode, a carbon paper-based catalyst-coated electrode as the cathode, and a solution consisting of 1M NaOH and 1M KCl as the electrolyte. The model integrates the dilute solution theory to explain species transport, the Euler-Euler (EE) multiphase approach to model the conversion of CO₂ from gaseous to aqueous phase, and the porous electrode theory to model the electrochemical conversion of H₂ within porous media.

The experimental data from Kim et al. [3] is used to validate the model developed. Using the proposed model, local electrolyte potential distribution, local electrolyte current distribution, local species concentration distribution, gas flow pattern, and time-dependent electrode current density, species concentration, and H₂ production are forecast. A parametric simulation is conducted on the flow rate, electrode potential and porosity. Herein, A novel Mg- CO₂ model is presented, designed for the continuous consumption of magnesium while generating Mg(HCO₃)₂, H₂, and electricity. This Mg- CO₂ model can be used to optimize CO₂ conversion and has the potential to promote environmental sustainability and make the world greener.

References

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